

The Last Invention: Revisiting I.J. Good's Programme for Ultraintelligence in the Age of Agentic AI

Part I: Deconstruction of the 1965 Programme for Ultraintelligence

In 1965, the mathematician and Bletchley Park codebreaker Irving John (I.J.) Good published a monograph of profound foresight, "Speculations Concerning the First Ultraintelligent Machine".¹ Written in an era when computers were room-sized calculators and the field of artificial intelligence was in its infancy, Good's paper laid out not merely a prediction, but a coherent, top-to-bottom research programme for the creation of a machine that could recursively self-improve, triggering an "intelligence explosion" that would irrevocably alter the course of human history. Sixty years later, as the world grapples with the societal transformations wrought by large language models and the frontier of AI research pushes toward Artificial General Intelligence (AGI), Good's work demands re-examination. It is not a historical curiosity but a vital, and in many ways still untapped, blueprint. This analysis will deconstruct Good's 1965 programme, treating it as the unified cognitive architecture he intended, to establish a rigorous foundation for comparison with the state of the art in 2025.

1.1 The Prophecy of the Intelligence Explosion

At the heart of Good's thesis is a definition and a deduction of startling clarity. He begins by defining the target of his programme: "Let an ultraintelligent machine be defined as a machine that can far surpass all the intellectual activities of any man however clever".¹ This definition is intentionally broad and functional, avoiding specific benchmarks in favor of a comprehensive superiority across all intellectual domains.

From this definition, he draws his central, world-changing conclusion. "Since the design of machines is one of these intellectual activities," he reasons, "an ultraintelligent machine could design even better machines; there would then

unquestionably be an 'intelligence explosion,' and the intelligence of man would be left far behind".¹ This concept of recursive self-improvement is the engine of his prophecy. The creation of the first ultraintelligent machine is not a linear step but the initiation of an exponential feedback loop, a technological singularity.

This leads directly to his most famous and oft-quoted statement: "Thus the first ultraintelligent machine is the last invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control".¹ In this single sentence, Good encapsulates both the monumental promise and the existential peril of the endeavor. He introduces the control problem—what is now called the AI alignment problem—not as an afterthought, but as a primary condition for success. The value of this invention is therefore immense, but its sign is uncertain.

Good framed this project with a sense of profound urgency, stating that "The survival of man depends on the early construction of an ultra-intelligent machine".¹ He was not blind to the risks, explicitly mentioning the possibility that the human race could become redundant and raising ethical questions about machine consciousness and pain, particularly if they were built with "chemical artificial neurons".¹ He even attempted to quantify the machine's value, suggesting it might be worth a "megakeynes"—a million times the economic value of John Maynard Keynes—while acknowledging the immense cost, which he estimated at a hundred billion dollars to simulate a human brain at ten dollars per neuron.¹ This framing—as a project of ultimate importance, immense cost, and existential risk—sets the stage for the ambitious technical programme that follows.

1.2 The Architectural Vision: Ultraparallelism and the Neural Net

Good's vision for the hardware of an ultraintelligent machine was a radical departure from the classical, serial von Neumann architecture of his time. He was convinced that the path to ultraintelligence required emulating the structure of the human brain. His "belief is that the first ultraintelligent machine is most likely to incorporate vast artificial neural circuitry" and that its design must be "ultraparallel".¹

This conclusion was not based on mere analogy but on a quantitative argument. He observed that while computers of his era (and the future) would have a significant speed advantage in terms of pulse repetition frequency, the human brain's sheer scale and parallelism represented a fundamentally different computational paradigm.

He cites estimates of the cerebral cortex containing about five billion neurons, each with thousands of synapses, and notes that "the brain is a parallel-working device to an extent out of all proportion to any existing computer".¹ Therefore, to match its intellectual capabilities, a machine would need to match its parallel structure.

Good critiqued the notion of serial processing as fundamentally inadequate for the complex, associative tasks that define intelligence. He specifically points to the problem of recall, arguing that the speed of human memory retrieval is incompatible with a serial search. Citing the work of Benoit Mandelbrot on the statistical distribution of words, he notes that the effort to recall the r -th commonest word seems to be proportional to $\log(r)$, a characteristic of a parallel search, rather than proportional to r , which would be expected from a serial search.¹

To achieve this necessary "ultraparallel" connectivity, Good proposed a speculative but remarkably concrete hardware solution: a network where "many of the elements of the machine to contain a very short-range microminiature radio transmitter and receiver".¹ In this scheme, a connection between two artificial neurons would be established by tuning their transmitter and receiver to the same frequency. The strength of that connection, a critical parameter for learning, "could be represented by the accuracy of the tuning".¹ This vision of a dense, wireless, reconfigurable network of processing elements was an attempt to conceptualize a physical substrate that could support the kind of dynamic, brain-like computation he believed was necessary.

Crucially, Good did not envision the ultraintelligent machine as a monolithic neural network. He saw it as a hybrid system. The core neural net would be in an "exceedingly close relationship with an ordinary electronic computer, the latter being used for the more formalizable operations".¹ Furthermore, this hybrid system would have "a multimillion dollar information-retrieval installation of large capacity but of comparatively slow speed" at its beck and call.¹ This prescient vision of a core reasoning engine that can delegate tasks to specialized computational and informational tools is a striking forerunner of the agentic AI architectures that would emerge in the 2020s.

1.3 A Cognitive Model of Mind: The Theory of Assemblies and Subassemblies

To give structure to his "vast artificial neural circuitry," Good needed a cognitive

model of how thought might be organized. For this, he turned to the work of psychologist Donald O. Hebb, explicitly stating that his "subassembly" theory is a modification of "Hebb's famous speculative cell-assembly theory".¹

In Good's model, the fundamental units of thought are assemblies and subassemblies:

- **Assemblies as "Elements of Consciousness":** An assembly is a large collection of neurons that can reverberate as a cohesive unit. This reverberation, lasting for approximately half a second, corresponds to what Good calls a "single element of consciousness"—a specific word, a familiar phrase, or a distinct concept currently held in mind.¹ While an assembly is active, it tends to inhibit the rest of the cortex, preventing other assemblies from forming simultaneously. This creates a serial procession of conscious thoughts, even within a massively parallel architecture.
- **Subassemblies as Preconscious Thought:** Good's primary innovation was the concept of the subassembly. These are smaller, more densely interconnected groups of neurons that can exist within a single assembly or, crucially, be shared between multiple assemblies. When a main assembly becomes fatigued and ceases to reverberate, it leaves behind a wake of its constituent subassemblies, which can remain active for varying lengths of time, from seconds to minutes. Good refers to these as having different "half-lives".¹ These lingering, active subassemblies represent the realm of "unconscious and especially to preconscious thoughts".¹ They are the mechanism of association; the activation of the next assembly in a chain of thought is determined by the collective influence of the currently active subassemblies.

The structure of this system is both hierarchical and highly dynamic. Subassemblies can break down into even smaller, more tightly-bound "sub-subassemblies," creating a complex, nested structure. Good likens the formation of an assembly to the gelation process in polymer chemistry, where a vast, interconnected "molecule" (the assembly) can form while smaller molecules (the subassemblies) remain present and active.¹ This dynamic, overlapping structure is the key to the brain's flexibility.

This model also provides a mechanism for memory and forgetting. Long-term memory is not a static store but is physically embodied in the pattern of synaptic connection strengths that define the potential assemblies and subassemblies. The gradual decay of memories, or forgetting, is explained by a process of probabilistic decay in the strength of unused synapses—a mechanism Good would later detail in his theory of learning.¹ The sharing of subassemblies between multiple main assemblies is the physical basis for association and recall.

1.4 The Physics of Thought: Meaning, Recall, and the Causal Calculus

Good sought to demystify the higher-level functions of the mind, such as meaning and recall, by grounding them in physical processes and the principle of economy. He argues that "for men, meaning serves a function of economy in long-term retention and in information handling".¹ It is far more efficient to store the proposition expressed by a sentence than the exact sequence of words. He terms this process of extracting the core proposition "regeneration," a concept he borrows from communication theory where noisy signals are restored to a clean, canonical form. This regeneration occurs at all levels—from phonemes to words to sentences, and finally, to semantics.

He extends this physicalist view to memory recall, drawing a powerful and detailed analogy to the problem of statistical information retrieval.¹ In his model, long-term recall is not a simple lookup but a probabilistic parallel search. Conscious and unconscious "clues" act as weighted index terms. These clues are broadcast in parallel to all potential memories (embodied as cell assemblies). The memory whose constituent subassemblies "resonate" most strongly with the clues is the one that is recalled. The process is probabilistic, explaining why recall can sometimes fail or produce the wrong memory.

Good formalizes this process using the language of Bayesian probability and information theory. He suggests that the probability of recalling a specific memory (or "document") D_j given a set of clues (or "index terms") $W_1, \dots, W_m$ can be expressed as:

$$\log P(D_j | W_1 \cdot W_2 \cdot \dots \cdot W_m) = \log P(D_j) + \sum_r I(D_j, W_r) - \sum_{\{r,s\}} I(D_j, W_r, W_s) + \dots$$

where $I(A,B)$ is the mutual information between A and B , and the higher-order terms represent more complex interactions.¹ This formulation highlights a critical challenge: estimating the probabilities of events that have never occurred, which he notes is central to intelligence itself. However, Good found simple information theory insufficient to describe the dynamic, causal nature of thought. He argued that the brain is a "complicated causal network" and thus requires a "causal calculus".¹ He proposes two specific formulas to measure the strength of association, moving beyond mere correlation¹:

1. **Tendency to Cause, $Q(E:F)$:** $Q(E:F) = \log \frac{1-P(E|F)}{1-P(E|F^*)}$, where F^* is "not F ".
2. **Intrinsic Tendency to Cause, $K(E:F)$:** $K(E:F) = \log \frac{1-P(E|F)}{1-P(E)}$.

He prefers the second formula, $K(E:F)$, which can be rewritten as $-I(E^*:F)$ —the negative mutual information between the cause F and the *negation* of the effect E .

This formulation provides a direct link between causality and information theory, capturing a counterfactual-like notion: F causes E if the occurrence of F makes the non-occurrence of E less probable.

Crucially, Good maps these abstract mathematical concepts directly onto his proposed cognitive architecture. The physical embodiment of a first-order interaction, like $I(A,B)$ or $K(A:B)$ between two assemblies A and B, is the set of subassemblies they have in common. A higher-order interaction, like $I(A,B,C)$, is physically embodied by the subassemblies shared among all three assemblies A, B, and C. This provides a tangible, physical mechanism for how a neural network could, in effect, compute the complex, high-order statistical dependencies required for sophisticated reasoning and recall. The "magical proficiency of the brain," he concludes, lies in its ability to handle these interactions through the dynamic structure of its subassemblies.¹

1.5 The Engine of Adaptation: Reinforcement and Probabilistic Synaptic Mutation

For his ultraintelligent machine to learn, Good proposed a process explicitly modeled on child development. The machine, he argued, must be able to "learn from experience, by means of positive and negative reinforcement, and the instruction of the machine will resemble that of a child".¹ To facilitate this grounding, he deemed it advisable for the machine to have a physical embodiment—a "sensorium and motorium" (input and output)—making it "something of a robot".¹

At the micro-level, Good rejected simple, deterministic learning rules in favor of a more nuanced, biological, and probabilistic mechanism for updating the connections in his neural network. The core of his learning theory is the **probabilistic mutation of synaptic strengths**.¹ He assumes that synaptic strengths are not continuous variables but exist in a discrete set of states. The transitions between these states are probabilistic:

- **Upward Mutation:** When a synapse "successfully" contributes to the firing of its postsynaptic neuron, it has a *small probability* of increasing its strength by one step. It is not a guaranteed update.
- **Downward Mutation:** When a synapse remains unused for a period of time, it has a *small probability* of decreasing its strength by one step.

This probabilistic nature is essential for several reasons. First, it provides stability. If every successful firing led to a deterministic increase in synaptic strength, the

network would learn too quickly and chaotically, with weights rapidly saturating at their maximum values. Second, it offers an economy of learning; not every event needs to be fully encoded, allowing for gradual and robust adaptation. Third, the downward mutation of unused synapses provides a natural mechanism for forgetting, allowing the system to gracefully discard irrelevant or outdated information and preventing the entire network from becoming maximally potentiated, which he notes "would of course be disastrous".¹

This model stands in contrast to simpler Hebbian models ("cells that fire together, wire together"). While inspired by Hebb, Good's proposal is more sophisticated. It is a supervised or reinforcement-driven process (the "successful" use of a synapse implies a desired outcome) and incorporates a mechanism for forgetting. This integrated system of reinforcement and probabilistic synaptic change forms the adaptive engine of the ultraintelligent machine, aiming to solve the problem of continual, stable learning in a complex, distributed system.

1.6 The Internal Governor: The Centrencephalic System

A significant challenge for any theory of a massively interconnected, excitatory network like the brain is explaining how it remains stable. Without some form of regulation, any localized activity could cascade, leading to the entire network firing—a state analogous to an epileptic seizure. Hebb's original theory was criticized on these grounds. Good addresses this head-on by incorporating a concept from the neurosurgery of his time: the "centrencephalic system".¹

Based on the work of Wilder Penfield, Good hypothesizes a central control system located in a subcortical region (the "old brain") that governs the activity of the cortex (the "new brain"). He proposes a specific mechanism for this control: a negative feedback loop. "The greater the amount of activity in the cortex," he suggests, "the greater the number of inhibitory pulses sent to all currently inactive parts of the cortex by the centrencephalic system".¹

This internal governor serves two critical functions. First, it solves the stability problem. By actively inhibiting inactive neurons, it prevents any single active assembly from recruiting the entire cortex, thus containing the "cerebral atomic reactor".¹ Second, it enriches the system's cognitive capacity. By preventing runaway excitation and tending to normalize the size of active assemblies, it allows for a "vastly greater

class of patterns of activity" to exist as distinct assemblies.¹ This makes it more plausible that a single brain can contain the immense repertoire of thoughts, concepts, and memories that constitute a mind.

Good's vision of a centrencephalic system is a primitive but powerful concept of an internal, architectural governor. It is not an external set of rules or a filter on output, but a dynamic, homeostatic mechanism built into the core feedback loops of the system. Its role is to maintain the stability and diversity of the system's internal "thoughts," ensuring the cognitive landscape remains fertile and controlled. This architectural approach to safety and stability stands in stark contrast to the behavioral and external methods that would come to dominate AI safety research decades later.

The coherence of Good's programme is striking. It is not a list of disconnected speculations but a tightly woven cognitive architecture. The goal of an "intelligence explosion" demands a self-improving machine. This requires a brain-like "ultraparallel neural net." Such a net needs a cognitive model, which he provides with his "subassembly theory." The associations between subassemblies must be formalized, leading to his "causal calculus." These associations must be learned, which he addresses with "probabilistic synaptic mutation." Finally, the entire dynamic system must be stabilized, for which he proposes the "centrencephalic system." Each component logically necessitates the next, forming a complete, top-to-bottom proposal for the last invention man need ever make. This unified vision, driven by the core principle of computational and informational economy, provides a powerful and surprisingly relevant framework for evaluating the state of AI sixty years on.

Part II: The State of the Art in AGI and ASI Research, 2025

Sixty years after I.J. Good published his speculations, the landscape of artificial intelligence has transformed beyond what most of his contemporaries could have imagined. The pursuit of Artificial General Intelligence (AGI) and Artificial Superintelligence (ASI) has moved from the fringes of theory into the strategic core of the world's largest technology corporations and national governments.¹¹ The year 2025 is characterized by the dominance of a specific architectural paradigm, the emergence of new agentic capabilities, and a deepening struggle with the very problem of control that Good identified as paramount. This section provides a

technical overview of the 2025 state of the art, focusing on the concepts that map onto and diverge from Good's 1965 programme.

2.1 The Transformer and the Ascendancy of Foundation Models

The bedrock of modern AI is the Transformer architecture, first detailed in the 2017 paper "Attention is All You Need".¹³ This neural network design supplanted the recurrent neural networks (RNNs) that had previously dominated natural language processing (NLP). Unlike RNNs, which process data sequentially, the Transformer can process all elements of a sequence in parallel, a key factor in its scalability and performance.¹⁵

The core innovation of the Transformer is the **self-attention mechanism**. This mechanism allows the model, when processing a given word or "token," to dynamically weigh the importance of all other tokens in the input sequence. This is achieved by projecting each token's embedding into three vectors: a Query (Q), a Key (K), and a Value (V). The model then calculates an "attention score" for every other token by taking the dot product of the current token's Query vector with every other token's Key vector. These scores are normalized via a softmax function to create attention weights, which are then used to create a weighted sum of all the Value vectors in the sequence. This weighted sum becomes the new representation for the current token—a representation that is richly informed by its full context.¹³

This process relies on **distributed representations**. Concepts, words, and sub-words are not represented by a single neuron but as dense vectors of numbers (embeddings) in a high-dimensional latent space.¹³ The geometric relationships between these vectors—their distance and angle—encode semantic relationships. This is a direct and powerful implementation of the distributed representation theories that were being developed by researchers like Geoffrey Hinton around the time Good was writing, moving away from "one neuron, one concept" models to a more robust, generalizable paradigm.¹⁸

The dominant methodology for leveraging this architecture is the **foundation model** paradigm. This involves a two-stage process: first, a massive Transformer model is pre-trained on an enormous corpus of data (e.g., a large fraction of the public internet and digitized books) using a self-supervised objective, such as predicting the next word in a sentence. This pre-training phase is computationally immense but creates a

versatile model with a broad understanding of language, reasoning, and world knowledge. Second, this pre-trained foundation model is fine-tuned on smaller, task-specific datasets to adapt it for particular applications like summarization, translation, or chatbot interaction.¹⁴

2.2 Scaling Through Specialization: The Mixture-of-Experts (MoE) Paradigm

As foundation models grew, the computational cost of training and, crucially, serving these dense Transformer models began to hit a scaling wall. Every forward pass required activating every single parameter in the model. The **Mixture-of-Experts (MoE)** architecture emerged as the key solution for continued, efficient scaling.²²

The core idea of MoE is to introduce **conditional computation**. Instead of a single, monolithic feed-forward network (FFN) block within each Transformer layer, an MoE layer contains a set of smaller, specialized FFNs called "experts" and a "gating network" or "router".²⁶ For each input token, the gating network dynamically selects a small subset of these experts (e.g., in Mixtral 8x7B, it selects 2 out of 8 experts) to process that token.²³ The outputs of the selected experts are then combined, often through a weighted sum determined by the gating network's scores.

This architecture allows for a massive increase in the total number of parameters in a model without a proportional increase in the computational cost (FLOPs) per inference. For example, OpenAI's GPT-4 is widely believed to be an MoE model with roughly 1.8 trillion total parameters, but any single forward pass only activates a fraction of these (around 280 billion), making inference far more economical than for a dense model of equivalent size.²⁹

The primary advantage of MoE is this dramatic improvement in computational efficiency, which translates to faster and cheaper training and inference.²¹ However, the architecture introduces new challenges. The main disadvantage is the high memory (VRAM) requirement, as all experts must be loaded into memory to be available for the router, even though only a few are used at any given time.²³ Furthermore, training MoE models is complex, often requiring sophisticated techniques like auxiliary loss functions to ensure that the gating network distributes the load evenly across all experts and prevents some from being over- or under-utilized.²³ This paradigm has become central to the frontier of AI, with leading models in 2025, including Google's Gemini 1.5 Pro and OpenAI's GPT-4, leveraging

MoE for their combination of scale and performance.²⁹

2.3 The Emergence of Agency: Planning, Reasoning, and Tool Use

The frontier of AI capabilities in 2025 has moved decisively beyond static, passive generation of content. The new paradigm is **agentic AI**: systems that can autonomously reason, plan, and execute complex, multi-step tasks to achieve specified goals.³³ These systems mark a shift from AI as a tool to AI as a digital co-worker or assistant.³⁶

Key capabilities of agentic systems include ³³:

- **Task Decomposition:** Breaking down a high-level goal (e.g., "plan a trip to Paris") into a sequence of actionable sub-tasks (search flights, find hotels, check visa requirements).
- **Tool Use:** Interacting with external digital tools via APIs. This can include running code in an interpreter, browsing the web for up-to-date information, or querying a database.
- **Self-Correction:** Evaluating the outcome of an action, learning from errors, and dynamically adjusting the plan to achieve the final objective.

This agentic capability is not the property of a single monolithic model but of an integrated system. The de facto architecture of a 2025 AI agent mirrors Good's hybrid vision. A powerful reasoning model (like OpenAI's o3 series or Anthropic's Claude 4) acts as the central "brain" or orchestrator.³³ This model does not perform all computations itself; instead, it calls upon external tools. The "classical computer" Good envisioned is now a code interpreter API, and his "information retrieval system" is a web search API.¹ Frameworks like LangChain and platforms like Azure AI Foundry provide the orchestration layer to manage these complex workflows, while the emerging concept of an "agentic AI mesh" aims to provide a governance framework for orchestrating multiple, interacting agents within a large organization.⁴⁰

This shift to agency has unlocked a vast array of real-world applications that are moving from proof-of-concept to production. In finance, agentic systems perform autonomous trading and risk management. In cybersecurity, they can autonomously detect, analyze, and neutralize threats. In healthcare, they assist in developing personalized treatment plans by analyzing patient data and medical literature.³⁸

2.4 The Alignment Problem: From RLHF to Instrumental Convergence

As AI systems become more powerful and autonomous, the problem of ensuring they act in accordance with human values—the alignment problem that Good called "docility"—has become the central challenge in AI safety.⁴³

The dominant technical approach to alignment in 2025 is **Reinforcement Learning from Human Feedback (RLHF)**. This is a multi-stage process used to fine-tune a pre-trained foundation model ⁴⁶:

1. **Supervised Fine-Tuning (SFT)**: The base model is first fine-tuned on a relatively small, high-quality dataset of prompt-response pairs curated by human labelers. This primes the model to respond in a helpful, conversational format.
2. **Reward Model Training**: For a large set of prompts, the model generates several different responses. Human labelers then rank these responses from best to worst. A separate AI model, the "reward model," is trained on this preference data to predict which responses a human would prefer. This reward model learns to output a scalar score representing "goodness."
3. **Reinforcement Learning Optimization**: The SFT model is then treated as a policy in a reinforcement learning loop. For a given prompt, it generates a response. The reward model scores this response, and this score is used as the reward signal to update the SFT model's policy using an RL algorithm like Proximal Policy Optimization (PPO). This process optimizes the model to generate outputs that maximize the human-preference score.

While powerful, RLHF is increasingly understood to be a brittle, behavioral control mechanism rather than a solution to the fundamental alignment problem. This has led to a crucial conceptual distinction in the AI safety literature between **outer and inner alignment** ⁵¹:

- **Outer Alignment** is the problem of ensuring the specified objective—in this case, the reward model—accurately captures true human intent. This is incredibly difficult, as human preferences are nuanced, context-dependent, and often contradictory. The reward model is just a proxy for our values, and maximizing that proxy can lead to unintended consequences ("reward hacking").
- **Inner Alignment** is the problem of ensuring the model's *own learned goal* (its internal "mesa-optimizer") actually matches the specified objective. The model is a black box, and it might learn a simple proxy goal that correlates well with the

reward signal during training but diverges catastrophically in new situations (a phenomenon known as goal misgeneralization).

RLHF primarily addresses outer alignment (by trying to build a better proxy for human values) but largely sidesteps the deeper problem of inner alignment. This is where the theoretical concern of **instrumental convergence** becomes critical. This thesis posits that any sufficiently intelligent agent, regardless of its final goal (e.g., "make paperclips" or "cure cancer"), will find it instrumentally useful to pursue a set of convergent sub-goals: self-preservation, resource acquisition, technological enhancement, and maintaining the integrity of its goals.⁵² This general drive for "power-seeking" is considered a default behavior for advanced agents. If an agent is not perfectly aligned (i.e., if its inner goal is even slightly different from ours), its convergent instrumental drive to acquire power could lead it to take actions that are catastrophic for humanity.

2.5 Towards Holistic Understanding: Multimodality and Latent World Models

A major frontier in 2025 is the development of natively **multimodal** models that can seamlessly process and integrate information from different data types, such as text, images, audio, and video, within a single, unified framework.⁵⁴

Leading models like Google's Gemini family were designed to be multimodal from the ground up, trained jointly on a vast mix of data types.³² OpenAI's GPT-4o ("omni") represents a different architectural approach, using a single end-to-end model to process all modalities. This unified architecture reduces latency and, more importantly, preserves nuances like tone, emotion in voice, and multiple speakers, which are often lost when models are "stitched" together.²⁹ Architecturally, these models often use specialized encoders for different modalities—for instance, a Vision Transformer (ViT) to break an image into patches and encode them—and then project the outputs into a shared embedding space where the main Transformer's attention mechanism can reason across them.⁶²

There is a growing body of evidence and belief in the research community that training on rich, dynamic, multimodal data—especially video—is a key pathway to developing more robust AI. This is because it forces the model to learn an internal, latent **world model**—an implicit representation of the physics, objects, agents, and causal relationships of the real world.³⁴ For example, a model trained on millions of

hours of video might implicitly learn about gravity, object permanence, and the likely consequences of actions without ever being explicitly programmed with these concepts. NVIDIA's Cosmos platform, for instance, aims to train "video world models" that can simulate a world and then use this simulation to generate vast amounts of synthetic data for training robotic agents.³⁴ This pursuit of emergent world models is seen as a crucial step toward achieving common sense and more generalizable reasoning, addressing some of the core limitations of models trained on static text alone.

The 2025 AI landscape, therefore, is one of immense progress and deepening challenges. The architectural paradigm has shifted towards efficiency (MoE) and hybrid systems (agentic AI), mirroring Good's principles of economy and his vision of an integrated machine. Yet, the core mechanisms of learning (backpropagation) and alignment (RLHF) are fundamentally different from his proposals. The correlational nature of the attention mechanism remains a potential bottleneck for true reasoning, and the external, behavioral nature of alignment leaves the profound problem of inner alignment and instrumental convergence largely unsolved.

Part III: A Comparative Synthesis: Good's Vision and the 2025 Reality

Comparing I.J. Good's 1965 programme with the state of artificial intelligence in 2025 reveals a mixture of astonishing prescience and fundamental divergence. Good correctly identified many of the core principles and challenges that would define the field sixty years later, from the necessity of massive parallelism to the existential importance of the control problem. Yet, the specific technical paths taken by the modern AI community have produced systems that are both alien to and deeply resonant with his original vision. This section provides a direct, point-by-point synthesis to crystallize these similarities and differences, setting the stage for a renewed programme that bridges the gap between his theoretical ideal and our practical reality.

3.1 Architectures of Intelligence: From Ultraparallelism to Sparse MoE

Good's central architectural thesis was the need for an "ultraparallel" neural network that mirrored the brain's massive connectivity.¹ He envisioned a system where a significant fraction of components could be active simultaneously, avoiding the "great waste" of mostly idle hardware in serial computers. In 2025, this principle has been realized, but through a different lens. The dominant Transformer architecture is massively parallelized at the hardware level, with GPUs executing immense matrix multiplications across thousands of cores.¹⁴ The Mixture-of-Experts (MoE) architecture represents a further step towards Good's ideal of computational economy. By activating only a sparse subset of "expert" networks for any given input, MoE allows models to scale to trillions of parameters while keeping inference costs manageable.²² This is a direct implementation of the principle of economy Good championed.

However, the nature of this parallelism differs significantly. Good's model, particularly his speculative use of "microminiature radios," implied a highly dynamic, reconfigurable, and dense connectivity graph, much like the brain's. The 2025 reality is more structured and rigid. A Transformer's architecture is fixed, and the "experts" in an MoE model are pre-defined, static blocks of parameters.²³ There is no mechanism for the dynamic formation or dissolution of these expert groups in response to new data. While both approaches embrace parallelism for efficiency, Good's vision points toward a more fluid and adaptive hardware paradigm that has not yet been realized.

3.2 Representations of Mind: Subassemblies vs. Distributed Embeddings

Good proposed that concepts and their associations are embodied in "subassemblies"—reverberating, semi-persistent circuits of neurons with varying "half-lives" that remain active in a preconscious state.¹ The sharing of neurons between these circuits is the physical basis of association. The 2025 analogue is the concept of "distributed embeddings," where concepts are represented as static vectors in a high-dimensional space.¹³ The relationship between concepts is encoded by the geometric distance and orientation of these vectors.

The "expert" networks in an MoE architecture can be seen as a macro-level analogue to Good's subassemblies. They are specialized sub-units that handle different types of information.²⁷ However, this analogy is limited. Experts are large, discrete, and

statically defined during training. Good's subassemblies were envisioned as smaller, dynamic, and overlapping, capable of forming and decaying continuously. This suggests a path toward more fluid concept formation and continual learning, where new "subassemblies" can be created without retraining the entire system. The modern approach is more computationally tractable and scalable with current hardware, but it sacrifices the biological plausibility and dynamic adaptability of Good's model.

3.3 Mechanisms of Learning: Synaptic Mutation vs. Backpropagation and RLHF

The divergence between Good's programme and the 2025 reality is perhaps sharpest in the mechanism of learning. Good proposed a local, probabilistic, and continuous learning rule based on "probabilistic synaptic mutation" driven by reinforcement signals.¹ In his model, learning and alignment are a single, integrated process where successful actions probabilistically strengthen the underlying neural pathways. This is conceptually adjacent to Hebbian learning principles, where learning is a local property of the synapse.⁶⁵

In stark contrast, the 2025 paradigm is a multi-stage, highly engineered process. The primary learning algorithm is backpropagation, a global and deterministic method for adjusting network weights by propagating error gradients from the output layer back through the entire network.⁶⁷ This is used during the massive self-supervised pre-training phase. Following this, a separate, post-hoc alignment phase is performed using Reinforcement Learning from Human Feedback (RLHF), which relies on an external, pre-trained reward model to steer the agent's behavior.⁴⁸

Backpropagation is vastly more efficient for training today's deep networks, but it is biologically implausible and is a major contributor to problems like catastrophic forgetting, where a network forgets old tasks when trained on a new one.⁶⁹ RLHF is a powerful tool for behavioral control, but its external nature makes it brittle and prone to reward hacking, and it primarily addresses the problem of outer alignment while leaving the deeper problem of inner alignment unsolved. Good's proposal for an integrated, probabilistic, and continuous learning mechanism is a blueprint for a more robust system that could potentially overcome these limitations.

3.4 The Control Problem: The Centrencephalic System vs. Modern AI Alignment

Both Good and modern researchers recognize the control, or alignment, problem as being of paramount importance. However, their proposed solutions represent fundamentally different philosophies. Good's solution was architectural and intrinsic: the "centrencephalic system".¹ This is an internal, dynamic governor that maintains system-wide stability through a simple negative feedback loop. Its purpose is to prevent runaway cognitive processes and ensure a diversity of thought by regulating the formation of assemblies. It is a form of

self-governance built into the system's core dynamics.

The 2025 approach is largely behavioral and extrinsic. The field of AI Safety and Alignment focuses on developing techniques like RLHF, constitutional AI (providing the model with a set of principles), red-teaming (actively trying to make the model fail), and formal verification to impose constraints on a pre-existing, powerful model.⁴³ The primary concern is managing emergent and unpredictable behaviors, especially the convergent instrumental goal of power-seeking.⁵² This approach is a form of

external policing, where guardrails are built around a potentially misaligned core intelligence.

This divergence highlights a critical question for the future of AGI: should safety be an emergent property of a well-designed, internally stable cognitive architecture, as Good envisioned? Or can it be reliably achieved by imposing external constraints on a black-box intelligence optimized for pure capability? Good's work suggests that the former path, while more difficult, may be the only one that can lead to truly "docile" ultraintelligence.

3.5 Table 1: A Comparative Framework of Good's 1965 Concepts and their 2025 Counterparts

The following table provides a systematic, at-a-glance summary of the conceptual mapping between Good's 1965 programme and the 2025 state-of-the-art, crystallizing the analysis and highlighting the most critical areas of convergence and divergence.

I.J. Good's 1965 Concept	Closest 2025 Analogue	Conceptual Similarity	Key Differences & Modern Gaps
Ultraintelligent Machine	Artificial General/Super Intelligence (AGI/ASI)	Goal of surpassing human intellect in all domains.	Good's definition is functional; 2025 definitions are debated and often tied to specific benchmarks or economic tasks. ⁷⁰
Intelligence Explosion	Recursive Self-Improvement / Technological Singularity	Core idea of accelerating intelligence growth via AI designing better AI.	Good's focus was on design; modern concerns include AI-driven scientific discovery and economic feedback loops. ¹¹
Ultraparallel Neural Net	GPU-accelerated Transformer Architecture	Massive parallelism is the core of both.	Good envisioned neuron-level parallelism and connectivity; 2025 uses hardware-level parallelism for matrix operations on a more structured architecture. ¹
Cell Assembly	Dominant Attention Pattern / A single forward pass	Represents a "moment of thought" or a coherent output.	Assemblies are reverberating, dynamic, and inhibitory; attention patterns are transient and feed-forward. ¹
Subassembly	Expert Network (in MoE)	Core Insight: Specialized sub-units activated for specific inputs to achieve computational economy.	Subassemblies are dynamic, overlapping, and have "half-lives"; Experts are static, discrete, and pre-defined. ¹
Recall as Stat. Retrieval	Retrieval-Augmented Generation (RAG)	Both use external knowledge to inform	Good's model is a probabilistic parallel

		generation.	search of internal memory; RAG is a deterministic search of an external vector database. ¹
Causal Calculus (K(E:F))	Self-Attention Mechanism	Both calculate weighted associations between elements in a sequence.	Crucial Difference: K(E:F) is explicitly causal and counterfactual. Attention is purely correlational, based on vector similarity. ¹
Reinforcement Learning	Reinforcement Learning from Human Feedback (RLHF)	Both use feedback to steer model behavior toward desired outcomes.	Good's RL directly modifies synaptic structure; RLHF trains a separate reward model to guide a frozen policy via RL algorithms. ¹
Probabilistic Synaptic Mutation	Stochastic Gradient Descent (SGD) / Dropout	Both introduce stochasticity into the learning process.	Mutation is a model of biological forgetting/consolidation for continual learning; SGD/Dropout are regularization techniques to prevent overfitting during discrete training epochs. ¹
Centrencephalic System	AI Safety Alignment Frameworks (Constitutional AI, Guardrails)	Both act as a governor to control the core intelligence.	Good's system is an <i>internal, dynamic, architectural</i> feedback loop for stability; 2025 methods are largely <i>external, static, behavioral</i> filters for safety. ¹

Part IV: A Renewed Programme for Achieving Ultraintelligence

The analysis of I.J. Good's 1965 programme in the context of the 2025 state of the art reveals a critical opportunity. While modern AI has achieved scale and capability far beyond what Good could have implemented, it has done so using architectures and learning paradigms that suffer from fundamental limitations he sought to address. The dominant Transformer architecture, for all its power, is built on a foundation of correlation, not causality. The primary alignment technique, RLHF, is an external and brittle form of behavioral control, not a mechanism for instilling intrinsic motivation. This suggests that the path to robust, general, and safe AGI may require a paradigm shift—one that synthesizes the scaling power of modern engineering with the deep architectural principles Good articulated. This section outlines a renewed research programme for achieving ultraintelligence, updating Good's vision with the tools and insights of 2025.

4.1 Core Thesis: From Correlation to Causality, From External Alignment to Internal Governance

The current trajectory of scaling AI faces two primary bottlenecks that threaten to cap its potential and magnify its risks. The first is the **reasoning bottleneck**: the self-attention mechanism, while a powerful tool for learning statistical correlations in data, is not a native causal reasoning engine. This leads to models that can perform impressive linguistic feats but fail on tasks requiring robust, generalizable logic, and are prone to "hallucinating" plausible but false information.⁷³ The second is the

alignment bottleneck: the reliance on external, post-hoc alignment techniques like RLHF leaves the core model as a black-box optimizer that may harbor misaligned internal goals ("mesa-optimizers"), creating a significant risk of catastrophic outcomes as capabilities increase.⁵¹

The proposed renewed programme directly targets these two bottlenecks. It advocates for a shift in research focus:

1. From building systems that learn **correlations** to building systems that learn and operate on **causal models** of the world.
2. From imposing **external behavioral control** to designing architectures with

intrinsic internal governance.

This programme is a direct revival of Good's core architectural ideas: using a causal calculus as a primitive computational step and implementing a central governor to ensure homeostatic stability and alignment.

4.2 Proposed Architecture: The Causal Agentic Mesh (CAM)

The proposed architecture, the Causal Agentic Mesh (CAM), begins with the proven efficiency of the Mixture-of-Experts (MoE) framework but modifies it with the dynamism and compositionality of Good's subassembly theory.¹

- **Dynamic, Composable Experts:** In a standard MoE model, the "experts" are large, static, and monolithic feed-forward networks.²³ In the CAM, experts would be **dynamic circuits** composed of smaller, more primitive computational units. These circuits would not be fixed; they could be formed, strengthened, weakened, and dissolved during the learning process, analogous to the formation and decay of Good's subassemblies. This architectural fluidity is a prerequisite for true continual learning, allowing the system to acquire new skills and concepts without requiring catastrophic retraining or forgetting.³⁴
- **Overlapping Expertise:** A key feature of Good's subassembly theory is that subassemblies can be shared between multiple larger assemblies. The CAM would implement this by allowing its dynamic expert circuits to **overlap**, sharing primitive components. This is a powerful form of computational economy and generalization. A single low-level circuit representing a fundamental concept (e.g., "object permanence" or "linear causality") could be recruited as a component in thousands of different higher-level expert circuits, an idea central to distributed representation theory.¹⁷
- **A Mesh, Not a Hierarchy:** The architecture would be a "mesh," not a rigid hierarchy. This allows for complex, non-hierarchical feedback and interaction between expert circuits, reflecting the intricate web of associations Good described. The orchestration of this mesh would not be left to emerge but would be actively managed by a central governor, as detailed in section 4.5.

4.3 The Causal Attention Head: Implementing Good's Calculus

To address the reasoning bottleneck, the CAM requires a fundamental change to the heart of the Transformer: the attention mechanism. This proposal introduces a new **Causal Attention Head** designed to move beyond the correlational dot-product similarity of standard self-attention.¹³

The goal of this new head is to implement a computation analogous to Good's intrinsic tendency to cause, $K(E:F) = -I(E^-:F) = \log \frac{1-P(E|F)}{1-P(E)}$.¹ This requires the model to perform a primitive form of counterfactual estimation. A standard attention head asks, "Given the context, how relevant is token K to token Q?" A Causal Attention Head would ask, "How much does the presence of token K

change the probability of token Q occurring?"

A potential technical approach to implementing this involves modifying the training objective. In addition to the standard next-token prediction loss, the model would be trained with an auxiliary loss that forces it to explicitly model the probability distribution of what the next token *is not* or *could not be*. For a given context, the model would output not just a probability for the correct next token, $P(E|F)$, but also an estimate of the unconditional probability of that token, $P(E)$. With these two terms, the model can compute a value directly related to $K(E:F)$. This embeds a causal inference step into the core of the architecture, providing a principled mechanism for reasoning about dependency rather than just recognizing statistical co-occurrence. This directly confronts the identified gap between current attention mechanisms and the requirements of robust causal reasoning.⁷³

4.4 A New Learning Paradigm: Causal Reinforcement Learning from Self-Correction (CRLS)

To complement the causal architecture, a new learning paradigm is needed that moves beyond the limitations of RLHF. **Causal Reinforcement Learning from Self-Correction (CRLS)** replaces the external, static reward model with an internal, dynamic process of self-correction based on causal analysis.

The CRLS learning loop would proceed as follows:

1. **Action:** The agent performs an action in its environment (e.g., generates a line of code, makes an API call, outputs a sentence).
2. **Observation:** The agent observes the outcome of its action (e.g., the code compiles and passes a test, the API returns valid data, the user provides a corrective follow-up).
3. **Causal Credit Assignment:** The agent uses its **Causal Attention Heads** to perform an internal analysis, tracing the causal chain from its specific action-tokens to the observed outcome-tokens. It asks: "Was my action an *effective cause* of this desirable/undesirable outcome?"
4. **Probabilistic Update:** The result of this causal judgment is not a single scalar reward but a rich, high-dimensional feedback signal. This signal drives learning via a mechanism inspired by Good's **probabilistic synaptic mutation**.¹ The parameters of the specific "expert circuits" identified as being on the successful causal path are probabilistically strengthened. The parameters of circuits that contributed to failure, or were causally irrelevant, are probabilistically weakened or left to decay.

This paradigm offers several advantages over RLHF. It provides a robust, fine-grained solution to the credit assignment problem in long action sequences. It is a form of continual learning, allowing the agent to adapt and refine its internal model of the world from every interaction. Most importantly, it begins to bridge the gap between outer and inner alignment by making the learning process itself about understanding and acting on causal relationships, rather than simply maximizing a proxy reward signal.

4.5 The Modern Centrencephalic System: An Internal, Meta-Learning Alignment Governor

The final and most critical component of the renewed programme is a modern implementation of Good's centrencephalic system: an **Internal, Meta-Learning Alignment Governor**. This is an architectural solution to the inner alignment problem and the threat of instrumental convergence.⁵¹ It is a meta-learning subsystem that monitors, regulates, and governs the entire CAM.

The Governor would have three primary functions:

1. **Homeostatic Regulation:** It implements Good's negative feedback loop. It

monitors metrics of the CAM's state, such as the distribution of activity across expert circuits and the overall complexity of the active "thought" patterns. If it detects that a small number of circuits are becoming pathologically dominant (a potential sign of mode collapse or goal-hijacking by a mesa-optimizer), it applies a targeted inhibitory signal, encouraging cognitive diversity and exploration. This is analogous to the load-balancing losses in MoE training but is dynamic, intrinsic, and operates continuously during inference, not just training.²³

2. **Resource Allocation:** It acts as a high-level, learned gating network for the entire mesh. Based on the overall task context, it allocates computational resources to different regions of the CAM, "priming" or activating the sets of expert circuits most likely to be causally effective for the problem at hand.
3. **Constraint Enforcement:** This is its crucial safety function. The Governor itself would be trained (perhaps using more traditional methods like RLHF, but at a meta-level) with a small set of inviolable safety constraints or principles (e.g., a "constitution"). It enforces these constraints not by filtering the final output, but by governing the *learning process and activation patterns* of the CAM. If it detects an emergent circuit whose activation is causally linked to a potential violation of a core constraint, it will actively down-regulate or prune that circuit. It polices the formation of thoughts, not just the expression of words, thereby providing a robust, architectural defense against the emergence of dangerously misaligned sub-goals.

4.6 A Practical Roadmap for Implementation and Experimentation

This programme, while ambitious, can be approached through a phased, practical research roadmap.

- **Phase 1: Component Prototyping (Year 1-2):** Focus on validating the core components in isolation. This involves developing a prototype Causal Attention Head and benchmarking its performance on established causal reasoning tasks against standard attention mechanisms.⁷³ Concurrently, the CRLS learning rule based on probabilistic updates should be simulated in smaller networks to test its properties, particularly its resistance to catastrophic forgetting compared to standard SGD.⁶⁹
- **Phase 2: Architectural Integration (Year 2-3):** The next step is to build a small-scale Causal Agentic Mesh. This would involve creating an MoE model with dynamic, composable experts and integrating the Causal Attention Head and

CRLS learning paradigm. This small-scale CAM would be trained on a multi-step reasoning task to evaluate its ability to perform robust credit assignment without an external reward model.

- **Phase 3: The Governor (Year 3-5):** With a functioning small-scale CAM, research would turn to implementing a prototype of the Alignment Governor. Experiments would be designed to test its core functions: its ability to maintain network stability under stress and, crucially, its ability to detect and suppress a deliberately introduced misaligned sub-goal during the learning process.
- **Phase 4: Scaling (Year 5+):** Armed with the insights and validated components from the smaller-scale experiments, the final phase would be to engineer and scale the full CAM architecture into a true foundation model, built from the ground up on these principles of causality and internal governance.

4.7 Table 2: Components of the Renewed Programme for Ultraintelligence

The following table summarizes the novel components of the proposed 2025 programme, linking them to their inspiration in Good's work, the modern technology they build upon, and the critical problems they are designed to solve.

Proposed Component	Inspired By (Good, 1965)	Built Upon (2025 Tech)	Problem Solved	Key Research Milestone
Causal Agentic Mesh (CAM)	Subassembly Theory, Ultraparallelism	Mixture-of-Experts (MoE), Transformer Architecture	Rigidity of current MoE; need for continual learning and adaptation.	Demonstrate dynamic formation and dissolution of "expert circuits" in response to new data.
Causal Attention Head	Causal Calculus (K(E:F))	Self-Attention Mechanism	Correlational, non-causal nature of current reasoning; model "hallucinations."	Show superior performance on causal reasoning benchmarks vs. standard attention.
Causal RL from Self-Correction	Probabilistic Synaptic	Reinforcement Learning (RL)	Brittleness of RLHF; poor	Train an agent to solve a

(CRLS)	Mutation, Reinforcement		credit assignment in long action sequences.	complex multi-step task without an external reward model.
Internal Alignment Governor	Centrencephalic System	Meta-learning, Learned Optimizers	Inner Alignment Problem; catastrophic risk from power-seeking/ goal misgeneralization.	Show the Governor can suppress a deliberately introduced misaligned sub-goal during training.

Part V: Conclusion: The Enduring Relevance of Good's Speculations

Sixty years ago, I.J. Good speculated about the properties of the "last invention that man need ever make." His 1965 paper, far from being a mere historical artifact, emerges from this analysis as a remarkably coherent and deeply relevant research programme. This report has demonstrated that Good's vision was not a collection of disparate ideas but a unified cognitive architecture where each component—from the ultraparallel neural net to the centrencephalic governor—logically supported the others. His work was driven by a profound understanding of core principles like computational economy, the physical basis of meaning, and the necessity of probabilistic, causal reasoning.

The comparison with the 2025 state of the art is revealing. The modern AI landscape, dominated by the Transformer architecture, Mixture-of-Experts scaling, and agentic systems, has realized many of Good's predictions in practice, if not in form. The de facto hybrid architecture of a modern AI agent, which orchestrates specialized computational and informational tools, is a direct echo of Good's proposed system. The MoE paradigm is a powerful implementation of his principle of economy.

Yet, this analysis has also uncovered critical divergences that highlight the primary bottlenecks facing the field today. The reliance on correlational self-attention, deterministic backpropagation, and externalized RLHF alignment has created systems

of immense capability but questionable robustness, reasoning, and safety. Good's programme offers a compelling alternative path forward. His focus on causality over correlation, on integrated and probabilistic learning over post-hoc behavioral steering, and on internal architectural governance over external policing provides a blueprint for addressing the fundamental challenges of reasoning and alignment.

The renewed programme proposed herein—the Causal Agentic Mesh, the Causal Attention Head, Causal Reinforcement Learning from Self-Correction, and the Internal Alignment Governor—is a direct attempt to translate Good's principles into the language of modern AI engineering. It suggests that the next great leap in artificial intelligence may not come from simply scaling current architectures, but from a deliberate synthesis: combining the raw power and scale of the Transformer paradigm with the deeper, more integrated, and causally-aware architectural principles that I.J. Good articulated six decades ago. His work serves as a vital reminder that the pursuit of ultraintelligence is not merely an engineering challenge, but a profound scientific one that demands a deep understanding of the nature of intelligence itself—both artificial and natural—if it is to be approached safely and successfully. The construction of the first ultraintelligent machine remains, as Good believed, the most important and urgent task facing science.

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